

LEARN PYTHON PROGRAMMING – BEGINNER TO MASTER

Section: 4. Operators and Expressions

Lecture: 31. Taking Input from Keyboard

Section 1: Lecture Summary

This lecture focuses on how to take input from the keyboard in Python programs, using the `input()` function. It builds on a previous program that calculates the area of a rectangle with fixed values for length and breadth. The main idea is to replace hardcoded values with dynamic user input, allowing the program to compute the area for different dimensions entered at runtime. The instructor explains the necessity of converting input (which is string by default) to numerical types like `int` or `float` using type conversion functions. The lecture demonstrates this through the example of reading length and breadth from the keyboard, calculating the area, and printing the result. It also covers how to run this program both in Jupyter Notebook and in other IDEs like PyCharm.

Section 2: Key Concepts and Explanations

- Need for Keyboard Input

- Fixed values limit flexibility.
- Taking input allows computation with varying data without modifying code.

- Python `input()` Function

- Built-in function to take input from the keyboard.
- Syntax: `variable = input('Prompt message')`
- Displays a prompt message to the user.
- Always returns data as a string.

- Type Conversion

- Input data from `input()` is always a string.
- For numeric calculations, convert strings to `int` or `float`.
- Use `int()` to convert to integer.
- Use `float()` to convert to floating-point number.
- Example: `length = int(input('Enter Length: '))`

- Example Program: Area of Rectangle with Input

- Read length and breadth from user input with prompts.
- Convert these inputs to integers.
- Calculate area by multiplying length and breadth.
- Print the area with a descriptive message.

- Running the Program

- In Jupyter Notebook, input prompts work but values can also be set directly for quick testing.
- In IDEs (PyCharm, VSCode), input is taken through run or terminal window.
- Program can be run multiple times with different inputs without changing the code.

Section 3: Example Code and Use Cases

Program to find the area of a rectangle using input from keyboard

```
length = int(input('Enter Length: '))
breadth = int(input('Enter Breadth: '))
area = length * breadth
print('Area:', area)
```

Explanation:

This program first prompts the user to enter the length and breadth of the rectangle. The inputs are converted from string to integer using `int()`, then multiplied to calculate the area. Finally, it prints the area with a label.

Section 4: Key Takeaways

- The `input()` function is used to read data from the keyboard.
- Input received using `input()` is always a string; numeric inputs require explicit conversion using `int()` or `float()`.
- Using keyboard input allows programs to process different data sets without changing the program code.
- The program to calculate the area of a rectangle can be generalized to accept any length and breadth entered by the user.
- Input operations work in interactive environments like Jupyter Notebook and in IDEs via the console.
- Practice writing programs that use `input()` to dynamically handle user data for flexible computations.